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ACETAN TABLET

(losartan potassium)

I. DESCRIPTION

Acetan Tablet 50mg : An ellipse, white film coated tablet plain on one side and '50' on the other side. Each tablet contains losartan potassium 50mg.
Acetan Tablet 100mg : An ellipse, white film coated tablet marked '100' on one side and scored on other side. Each tablet contains losartan potassium 100mg.

II. CLINICAL PHARMACOLOGY

Pharmacodynamic properties

Losartan is an oral angiotensin II receptor (type AT1) antagonist. Angiotensin II binds to the AT1 receptor found in many tissues (e.g. vascular smooth muscle, adrenal gland, kidneys and the heart) and elicits several important biological actions, including vasoconstriction and the release of aldosterone. Angiotensin II also stimulates smooth muscle cell proliferation. Based on binding and pharmacological bioassays, losartan binds selectively to the AT1 receptor. There is also an AT2 receptor found in many tissues. The functions of AT2 receptors have not been established. In vitro and in vivo, both losartan and its pharmacologically active carboxylic acid metabolite (E-3174) block all physiologically relevant actions of angiotensin II, regardless of the source or route of synthesis. The active metabolite is 10 to 40 times more potent by weight than losartan and appears to be a reversible, non-competitive inhibitor of the AT1 receptor.

During losartan administration, removal of angiotensin II negative feedback on renin secretion leads to increased plasma renin activity. Increases in plasma renin activity lead to increases in angiotensin II in plasma. Even with these increases, antihypertensive activity and suppression of plasma aldosterone concentration are maintained, indicating effective angiotensin II receptor blockade.

Losartan binds selectively to the AT1 receptor and does not bind to or block other hormone receptors or ion channels important in cardiovascular regulation. Furthermore, losartan does not inhibit ACE (kininase II), the enzyme that degrades bradykinin. Consequently, effects not directly related to blocking the AT1 receptor, such as the potentiation of bradykin mediated effects or the generation of oedema are not associated with losartan.

Losartan has been shown to block responses to angiotensin I and angiotensin II without affecting responses to bradykinin, a finding which is consistent with the specific mechanism of action of losartan. In contrast, ACE inhibitors have been shown to block responses to angiotensin I and enhance responses to bradykinin without altering the response to angiotensin II, thus providing a pharmacodynamic distinction between losartan and ACE inhibitors.

In non diabetic hypertensive patients with proteinuria, the administration of losartan potassium significantly reduces proteinuria, fractional excretion of albumin and IgG. Losartan maintains glomerular filtration rate and reduces filtration fraction. The mechanism of action of the uricosuric effect of losartan studied in normotensive subjects appears to be independent of angiotensin II blockade. Generally, in clinical trials, losartan caused a decrease in serum uric acid (usually 0.4mg/dL) which was persistent in chronic therapy.

Losartan has no effect on autonomic reflexes and no sustained effect on plasma noradrenaline. Losartan potassium, administered in doses of up to 150mg once daily, did not cause clinically important changes in fasting triglycerides, total cholesterol in patients with hypertension. The same doses of losartan had no effect on fasting glucose levels.

In patients with left ventricular failure, 25-mg and 50-mg doses of losartan produced positive haemodynamic and neurohormonal effects characterised by an increase in cardiac index and decreases in pulmonary capillary wedge pressure, systemic vascular resistance, mean systemic arterial pressure and heart rate and a reduction in circulating levels of aldosterone and noradrenaline. The occurrence of hypertension was dose related in these heart failure patients.

Pharmacokinetic properties

Absorption

Following oral administration, losartan is well absorbed and undergoes first-pass metabolism, forming an active carboxylic acid metabolite and other inactive metabolites. The systemic bioavailability of losartan tablets is approximately 33%. Mean peak concentrations of losartan and its active metabolite are reached in 1 hour and in 3-4 hours, respectively. There was no clinically significant effect on the plasma concentration profile of losartan when the drug was administered with a standardized meal.

Distribution

Both losartan and its active metabolite are ≥99% bound to plasma proteins, primarily albumin. The volume of distribution of losartan is 34 liters. Studies in rats indicate that losartan crosses the blood-brain barrier poorly, if at all.

Biotransformation

About 14% an intravenously-or orally-administered dose of losartan is converted to its active metabolite. Following oral and intravenous administration of C-labeled losartan potassium, circulating plasma radioactivity primarily is attributed to losartan and its active metabolite.

Minimal conversion of losartan to its active metabolite was seen in about one percent of individuals studied. In addition to the active metabolite, inactive metabolites are formed, including two major metabolites formed by hydroxylation of the butyl side chain and minor metabolite, an N-2 tetrazole glucuronide.

Elimination

Plasma clearance of losartan and its active metabolite is about 600 mL/min and 50 mL/min, respectively. Renal clearance of losartan and its active metabolite is

about 74 mL/min and 26mL/min, respectively. When losartan is administered orally, about 4% of the dose is excreted unchanged in the urine, and about 6% of the dose is excreted in the urine as active metabolite. The pharmacokinetics of losartan and its active metabolite are linear with oral losartan potassium doses up to 200 mg.

Following oral administration, plasma concentrations of losartan and its active metabolite decline polyexponentially with a terminal half-life of about 2 hours and hours, respectively. During once-daily dosing with 100 mg, neither losartan nor its active metabolite accumulates significantly in plasma.

Both biliary and urinary excretion contribute to the elimination of losartan and its metabolites. Following an oral dose of ¹⁴C-labeled losartan in man, about 35% of radioactivity is recovered in the urine and 58% in the feces.

Characteristics in Patients

Following oral administration in patient with mild to moderate alcoholic cirrhosis of the liver, plasma concentrations of losartan and its active metabolite were, respectively, 5-fold and 1.7-fold greater than those seen in young male volunteers. Neither losartan nor the active metabolite can be removed by hemodialysis.

Pharmacokinetics in paediatric patients

Pediatric Use

The pharmacokinetics of losartan have been investigated in 50 hypertensive pediatric patients >1 month to <16 years of age following once daily oral administration of approximately 0.54 to 0.77 mg/kg of losartan (mean doses). The active metabolite is formed from losartan in all age groups. Pharmacokinetics of losartan and its active metabolite are generally similar across the studied age groups and consistent with pharmacokinetic historic data in adults.

III. INDICATIONS

Hypertension

ACETAN is indicated for the treatment of hypertension.

Hypertensive Patients with Left Ventricular Hypertrophy

ACETAN is indicated to reduce the risk of stroke in patients with hypertension and left ventricular hypertrophy, but there is evidence that this benefit does not apply to Black patients (see RACE).

Nephropathy in Type 2 Diabetic Patients

Indicated for the treatment of diabetic nephropathy with an elevated serum creatinine and proteinuria (urinary albumin to creatinine ratio ≥300mg/g in patients with type 2 diabetes and a history of hypertension. In this population, ACETAN reduces the rate of progression of nephropathy as measured by the occurrence of doubling the serum creatinine or end stage renal disease (need for dialysis or renal transplantation) or death.

IV.DOSAGE AND ADMINISTRATION

ACETAN may be administered with or without food.

ACETAN may be administered with other antihypertensive agents.

Route of administration: Oral

Hypertension

The usual starting and maintenance dose is 50 mg once daily for most patients. The maximal antihypertensive effect is attained 3-6 weeks after initiation of therapy. Some patients may receive an additional benefit by increasing the dose to 100 mg once daily.

For patients with intravascular volume-depletion (e.g., those treated with high-dose diuretics), a starting dose of 25 mg once daily should be considered (see PRECAUTIONS).

No initial dosage adjustment is necessary for elderly patients with renal impairment, including patients on dialysis. A lower dose should be considered for patients with history of hepatic impairment (see PRECAUTIONS).

Hypertensive Patients with Left Ventricular Hypertrophy

The usual starting dose is 50 mg once daily. A low dose of hydrochlorothiazide should be added and/or the dose of ACETAN should be increased to 100 mg once daily based on blood pressure response.

Renal Protection in Type 2 Diabetic Patients with Proteinuria and Hypertension

The usual starting dose is 50 mg once daily. The dose may be increased to 100 mg once daily based on blood pressure response. ACETAN may be administered with other antihypertensive agents (e.g., diuretics, calcium channel blockers, alpha- or beta-blockers, and centrally acting agents) as well as with insulin and other commonly used hypoglycemic agents (e.g., sulfonylureas, glitazones and glucosidase inhibitors).

Pediatric Use

Neonates with a history of in utero exposure to ACETAN:

If oliguria or hypotension occur, direct attention toward support of blood pressure and renal perfusion. Exchange transfusions or dialysis may be required as a means of reversing hypotension and/or substituting for disordered renal function.

Antihypertensive effects of ACETAN have been established in hypertensive pediatric patients aged > 1 month to 16 years. Use of ACETAN in these age groups is supported by evidence from adequate and well-controlled studies of ACETAN in pediatric and adult patients as well as by literature in pediatric patients.

For patients who can swallow tablets, the recommended dose is 25 mg once daily in patients ≥20 to <50kg. The dose can be increased to a maximum of 50 mg once daily. In patients >50kg, the starting dose is 50 mg once daily. The dose can be increased to a maximum of 100 mg once daily.

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In pediatric patients who are intravascularly volume depleted, these conditions should be corrected prior to administration of ACETAN.

ACETAN is not recommended in pediatric patients with glomerular filtration rate <30 mL/min/1.73 m², in pediatric patients with hepatic impairment, or in neonates as no data are available.

V. CONTRAINDICATIONS

Concomitant use of losartan with aliskiren-containing products is contraindicated in patients with diabetes mellitus or renal impairment (GFR < 60 mL/min/1.73 m²)

VI. PRECAUTIONS

Hypersensitivity: Angioedema (see SIDE EFFECTS).

Hypotension and Electrolyte/Fluid Imbalance

In patients who are intravascularly volume-depleted (e.g., those treated with high-dose diuretics), symptomatic hypotension may occur. These conditions should be corrected prior to administration of ACETAN or a lower starting dose should be used (see DOSAGE AND 14 ADMINISTRATION).

Electrolyte imbalances are common in patients with renal impairment, with or without diabetes, and should be addressed,.

Liver Function Impairment

Based on pharmacokinetic data which demonstrate significantly increased plasma concentrations of losartan in cirrhotic patients, a lower dose should be considered for patients with a history of hepatic impairment (see DOSAGE AND ADMINISTRATION and CLINICAL PHARMACOLOGY, *Pharmacokinetics*).

Renal Function Impairment

As a consequence of inhibiting the renin-angiotensin system, changes in renal function including renal failure have been reported in susceptible individual; these changes in renal function may be reversible upon discontinuation of therapy.

Other drugs that effect the renin-angiotensin system may increase blood urea and serum creatinine in patients with bilateral renal artery stenosis or stenosis of the artery to a solitary kidney. Similar effects have been reported with ACETAN; these changes in renal function maybe reversible upon discontinuation of therapy. Losartan is not recommended in children with glomerular filtration rate < 30 mL/min/1.73 m, as no data are available.

Dual blockade of the renin-angiotensin-aldosterone system (RAAS)

There is evidence that the concomitant use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren increases the risk of hypotension, hyperkalaemia and decreased renal function (including acute renal failure). Dual blockade of RAAS through the combined use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren is therefore not recommended (see Interactions). If dual blockade therapy is considered absolutely necessary, this should only occur under specialist supervision and subject to frequent close monitoring of renal function, electrolytes and blood pressure. ACE-inhibitors and angiotensin II receptor blockers should not be used concomitantly in patients with diabetic nephropathy.

VII. PREGNANCY

When used in pregnancy during the second and third trimesters, drugs that act directly on the renin-angiotensin system can cause injury and even death in the developing fetus. When pregnancy is detected, ACETAN should be discontinued as soon as possible.

Although there is no experience with the use of ACETAN in pregnant women, animal studies with losartan potassium have demonstrated fetal and neonatal injury and death, the mechanism of which is believed to be pharmacologically mediated through effect on the renin-angiotensin system. In humans, fetal renal perfusion, which is dependent upon the development of the renin-angiotensin system, begins in the second trimester; thus, risk to the fetus increases if ACETAN is administered during the second or third trimesters of pregnancy.

VIII. NURSING MOTHERS

It is not known whether losartan is excreted in human milk. Because many drugs are excreted in human milk and because of the potential for adverse effect on the nursing infant, a decision should be made whether to discontinue nursing or discontinue the drug, taking into account the importance of the drug to the mother.

IX. DRUG INTERACTIONS

As with other drugs that block angiotensin II or it effects, concomitant use of potassium-sparing diuretics (e.g., spironolactone, triamterene, amiloride), potassium supplements, or salt substitutes containing potassium may lead to increases in serum potassium.

As with other drugs which affect the excretion of sodium, lithium excretion may be reduced. Therefore, serum lithium levels should be monitored carefully if lithium salts are to be co-administered with angiotensin II receptor antagonists.

Non-steroidal anti-inflammatory drugs (NSAIDs) including selective cyclooxygenase-2 inhibitors (COX-2 inhibitors) may reduce the effect of diuretics and other antihypertensive drugs. Therefore the antihypertensive effect of angiotensin II receptor antagonists may be attenuated by NSAIDs including selective COX-2 inhibitors.

In some patients with compromised renal function who are being treated with non-steroidal anti-inflammatory drugs, including selective cyclooxygenase-2 inhibitors, the co-administration of angiotensin II receptor antagonists may result in a further deterioration of renal function. These effects are usually reversible.

The use of losartan with an ACE inhibitor may increase the risk of hyperkalaemia, hypotension, and syncope, particularly in patients with atherosclerotic disease or heart failure, or in diabetics who have end-organ damage. Such combinations should be reserved for selected cases with close monitoring of renal function.

X. SIDE EFFECTS

ACETAN has been found to be generally well tolerated in controlled clinical trials for hypertension; side effects have usually been mild and transient in nature and have not required discontinuation of therapy. The overall incidence of side effects reported with ACETAN was comparable to placebo.

ACETAN was generally well tolerated in a controlled clinical trial in hypertensive patients with left ventricular hypertrophy. The most common drug-related side effect were dizziness, asthenia/ fatigue, and vertigo.

ACETAN was generally well tolerated in a controlled clinical trial in type 2 diabetic patients with proteinuria. The most common drug-related side effect were asthenia/fatigue, dizziness, hypotension and hyperkalemia (see PRECAUTIONS, Hypotension and Electrolyte/Fluid Imbalance).

The adverse experience profile for pediatric patients appears similar to that seen in adult patients.

The following additional adverse reactions has been reported in post-marketing experience:

Hypersensitivity: Anaphylactic reaction, angioedema including swelling of the larynx and glottis causing airway obstruction and/or swelling of the face, lips, pharynx and/or tongue has been reported rarely in patients treated with losartan; some of these patients previously experienced angioedema with other drugs including ACE inhibitors. Vasculitis, including Henoch-Schoenlein purpura, has been reported rarely.

Gastrointestinal: Hepatitis (reported rarely), liver function abnormalities, vomiting.

General disorder and administration site condition: Malaise.

Hematologic: Anemia, thrombocytopenia (reported rarely).

Musculoskeletal: Myalgia, arthralgia.

Nervous System/Psychiatric: Migraine, dysgeusia.

Reproductive system and breast disorders: Erectile dysfunction/impotence.

Respiratory: Cough.

Skin: Urticaria, pruritus, erythroderma, photosensitivity

XI. OVERDOSAGE

Limited data are available in regard to overdosage in humans. The most likely manifestation

of overdosage would be hypotension and tachycardia; bradycardia could occur from parasympathetic (vagal) stimulation. If symptomatic hypotension should occur, supportive treatment should be instituted.

Neither losartan nor the active metabolite can be removed by hemodialysis.

XII. SHELF LIFE

Please refer to the expiry date on the outer carton.

XIII. STORAGE CONDITION

Store at temperature below 30°C. Protect from light and moisture.

Keep out of reach of children.

Jauhi daripada kanak-kanak

XIX. AVAILABILITY

ACETAN 50 mg and 100 mg tablet are available in blister packs of 30's.

XV. PRODUCT REGISTRATION HOLDER

Duopharma Marketing Sdn, Bhd,

Lot No. 2, 4, 6, 8 & 10, Jalan P/7, Section 13, Bangi Industrial Estate, 43650 Bandar Baru Bangi, Selangor, Malaysia.

XVI. MANUFACTURER

Duopharma Manufacturing (Bangi) Sdn, Bhd,

Lot No. 2 & 4, Jalan P/7, Section 13, Bangi Industrial Estate, 43650 Bandar Baru Bangi, Selangor, Malaysia.

Revision Date: November 2024

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